semi-structures Python Package

Material-aware 2D diagrams, process-flow models and true 3D solids

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Version 0.1.0

17 August 2026

Declaration of generative AI use

This project was prepared with the assistance of generative AI models (OpenAI's 5.6 Sol and Anthropic's Claude Fable 5). The tools were used to draft and restructure code and prose, to propose organization, to assist with figures and documentation and to copyedit. All scientific content, technical judgments and structural decisions are my own. I have reviewed and verified the code, text, figures and references, and I am solely responsible for the accuracy of the work. Consistent with current publishing guidance, the AI systems are not authors of this project: they cannot take responsibility for the content and cannot hold or assign copyright, and they are therefore not credited as authors or co-authors.

Contents

1 Introduction

Semiconductor structures can be difficult to illustrate well. A useful figure must show the geometry clearly, keep material identity consistent, carry its labels and still read at publication size. General drawing tools build the shapes but carry no semiconductor material language and no repeatable style, so related figures get redrawn by hand, drift in color and resist revision when a layer, dimension or viewpoint changes.

Motivation

`semi-structures` makes those figures reproducible. A structure is written as Python, so geometry, labels, camera and output size can be reviewed, versioned and regenerated rather than locked inside an edited image. A shared palette gives every material a stable identity, so the same silicon, dielectric, metal or alloy stays recognizable across logic, memory, optoelectronic, power-device and packaging figures.

TCAD packages already render devices in three dimensions with real physics behind them, but they are expensive, tied to a license and built around designs for modeling work rather than around images for a paper or a talk. This package is the small free alternative for drawing a structure yourself. It is not a simulator and not a CAD system. It draws what it is told to draw.

The work began as a weekend project and continues a longer effort to test what current generative-AI tools can do on real technical work. It should be read in that light.

What the package provides

A structure is authored once, as a process. The module `semi_structures.process` holds the fabrication verbs: deposit a blanket or patterned film, etch, fill, implant, strip and form a mesa. Each verb records an ordered operation and returns a stable reference to the feature it created, so a model reads as a run sheet rather than as a picture.

Three renderers draw that model, and the choice between them is automatic. The vector isometric engine writes exact vector PDF for axis-aligned films, etches, fills and mesas. The section renderer cuts the model at a plane, which suits layer-order figures and tall stacks. The solid engine builds real meshes and cuts them with manifold3d, which is what circular holes, annular shells, transparency and capped cutaways require. Forcing the vector engine onto geometry it cannot draw exactly raises an error instead of quietly losing an operation.

One material language runs through all three. Hue gives the chemical family and lightness tracks electron density, so pale fills leave the saturated colors free for beams, fields and annotation. Text is sized in points at the printed width in every renderer, and no size below 7 pt is accepted.

Choose	When
`semi_structures.process`	Default authoring interface for semiconductor device structures and process semantics.
`semi_structures.iso`	Automatic output for safe rectilinear topology, or direct use for abstract vector diagrams.
`semi_structures.solid`	Automatic output for topology changes and curved geometry. Direct use for backend tests and assemblies.

Device structures should be written as process operations, and the renderer primitives are for packaging assemblies, backend tests and deliberately abstract diagrams. Every caption says which of the two it is. A

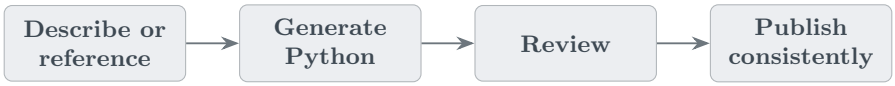
figure marked **Process DSL.** is authored as a fabrication sequence in the process domain-specific language (DSL), with its renderer chosen by dispatch, while one marked **Special purpose.** composes solids or Matplotlib content directly. Ten of the twenty-two figures are DSL based.

Installation, a quick start and worked usage are in `README.md`. The per-module verb tables, the label API and the palette reference are in `docs/guide.md`, and the command line and MCP server are in `docs/interfaces.md`. This document is a gallery, not a manual, and does not repeat them.

Assisted authoring, with judgment retained

The compact API is also suited to a generative-AI coding assistant, so an author who would rather describe a structure than construct it can work at that level and get an editable example back. A multimodal assistant can take a reference image for the arrangement, viewpoint or callouts. What comes back is a new schematic built from package primitives rather than a copy of the reference.

That is a starting point and not a substitute for review. The package does not infer a fabrication process from an image. The user remains responsible for dimensions, layer order, terminology, scientific accuracy and citation of reference material.



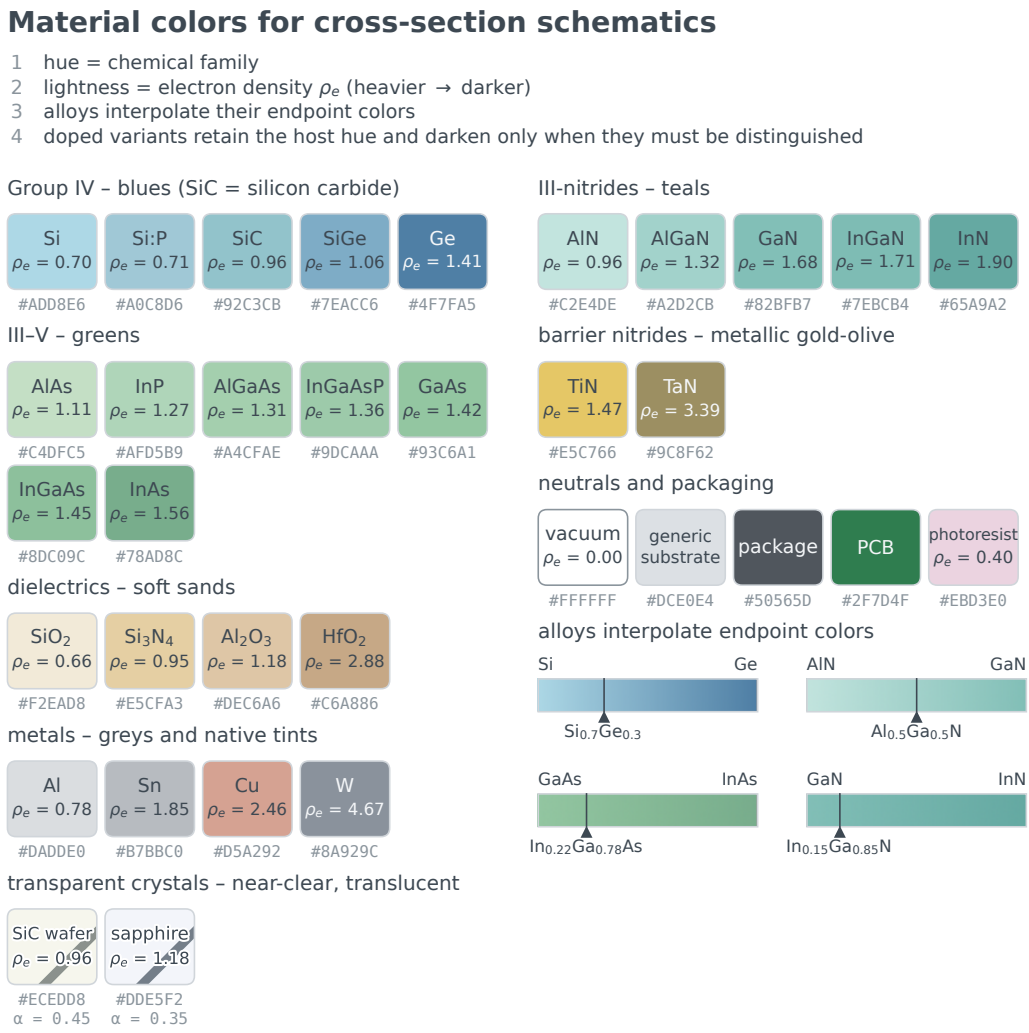
Status of this revision

The figures here are an early set. They show what the library produces quickly rather than what it produces at its best, and most would improve with another pass over camera choice and label placement. The mix of the two kinds is largely an accident of timing, because several special-purpose figures predate the process DSL and could be rebuilt as fabrication sequences today. Automatic label placement is the weakest part of the current output, since the two-column callout layout does not yet take account of the geometry underneath.

Every figure was made with the Python library, driven from a coding harness such as Claude Code or OpenAI Codex. None was produced through the MCP server or the command line. Those two interfaces are newer and lightly tested, so nothing in this gallery should be read as evidence about them.

2 Material language

The material key is the common contract between every renderer. Alloys are interpolated from their endpoint materials, and doped layers retain the host hue and move one shade darker only when they must be told apart from it, so a reader can identify a material by eye and not by consulting a legend. Packaging materials (package laminate, PCB, solder) are semantic entries with documented colors rather than an electron-density shade.



ρ_e: reference electron density (e Å^{-3}) from bulk density; higher values map to darker fills.
Substrate, package and PCB are semantic materials and take documented colors rather than a ρ_e shade.
Transparent crystals (bulk SiC wafer, sapphire = Al_2O_3) are near-clear tints drawn at the stated α by every renderer; the stripe shows the transparency.
They are distinct from the opaque device color SiC and the thin-film dielectric Al_2O_3.

Alloy ramps: x is the right-hand endpoint fraction; RGB interpolation is a drawing convention, not a physical model.

Figure 1: Special purpose. The semiconductor material palette: physical families on the left, nitrides, barriers, packaging materials and alloy ramps on the right, all on one chip grid. The last row on the left holds the transparent crystals, a bulk SiC wafer and sapphire (Al_2O_3). These are near-clear tints that every renderer draws at the stated alpha (the stripe behind each chip shows the transparency). They are distinct from the opaque device color SiC and the thin-film dielectric Al_2O_3. (Generated with `examples/fig_material_palette.py`).

3 Cross-sections and stack galleries

Layered structures are the simplest use case. The drawing code handles material lookup, alloy colors, repeated stacks, thickness annotations and consistent labeling, which leaves the geometry free to stay deliberately schematic. Six stacks that recur in metrology work are collected below, from a high-k metal gate to a W/Si multilayer mirror.

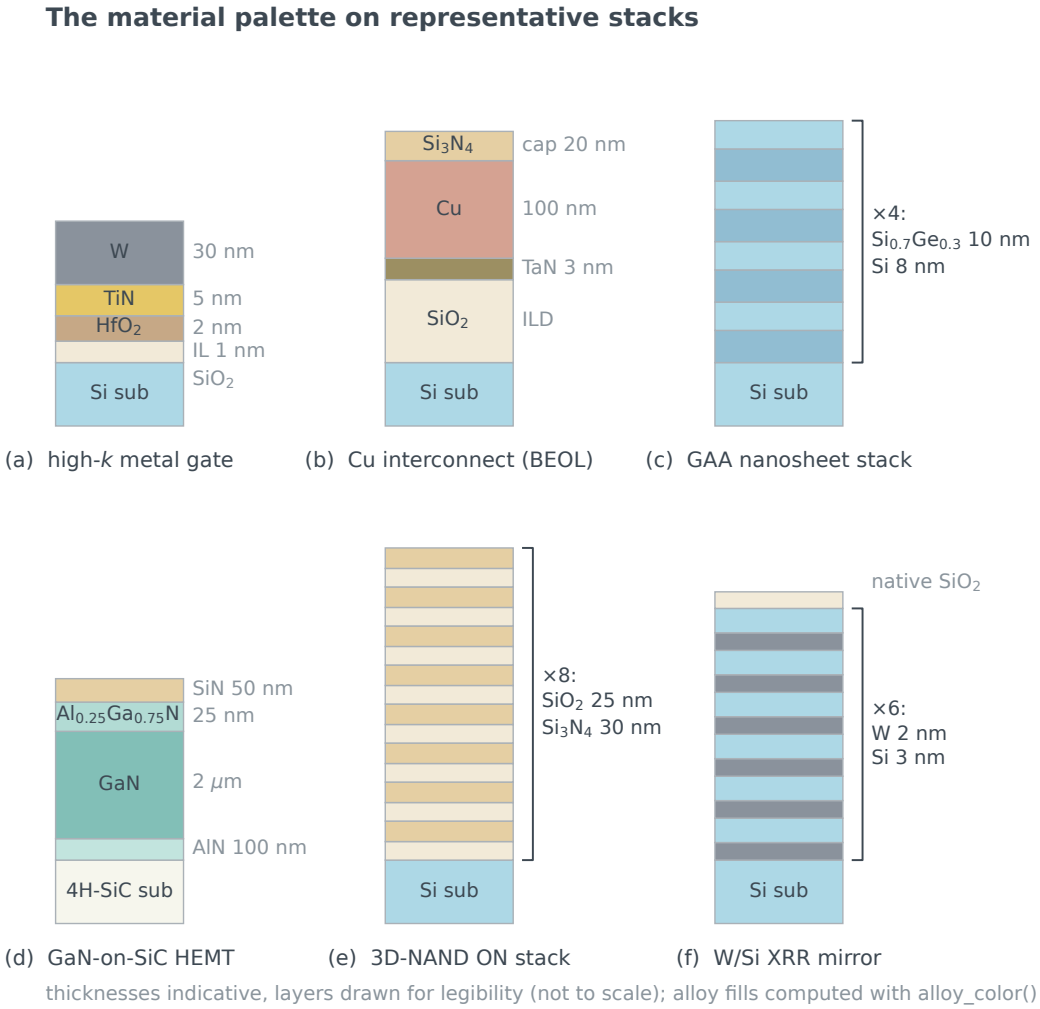


Figure 2: Special purpose. Representative metrology structures. (Generated with `examples/fig_stack_gallery.py`).

4 The transistor line-up in two renderings

One geometry, drawn twice, separates what the package fixes from what the author chooses. Both figures below are special-purpose compositions: their scripts place solids directly rather than running a fabrication sequence, because a line-up of four devices at one scale is a layout problem and not a process. Parallel projection preserves the technical-diagram look, and transparency exposes the channels inside a wrap-around gate.

Only the 3D panels are rasterized. Everything else in the saved PDF, including the labels, the leaders and the cross-sections, stays vector, so the page can be enlarged without softening the text.

The second rendering answers a request that comes up often. An author who needs the colors of a familiar diagram can have them, and the role scheme is then a deliberate exception to the palette law, written in a separate script so the material-colored figures stay the default.

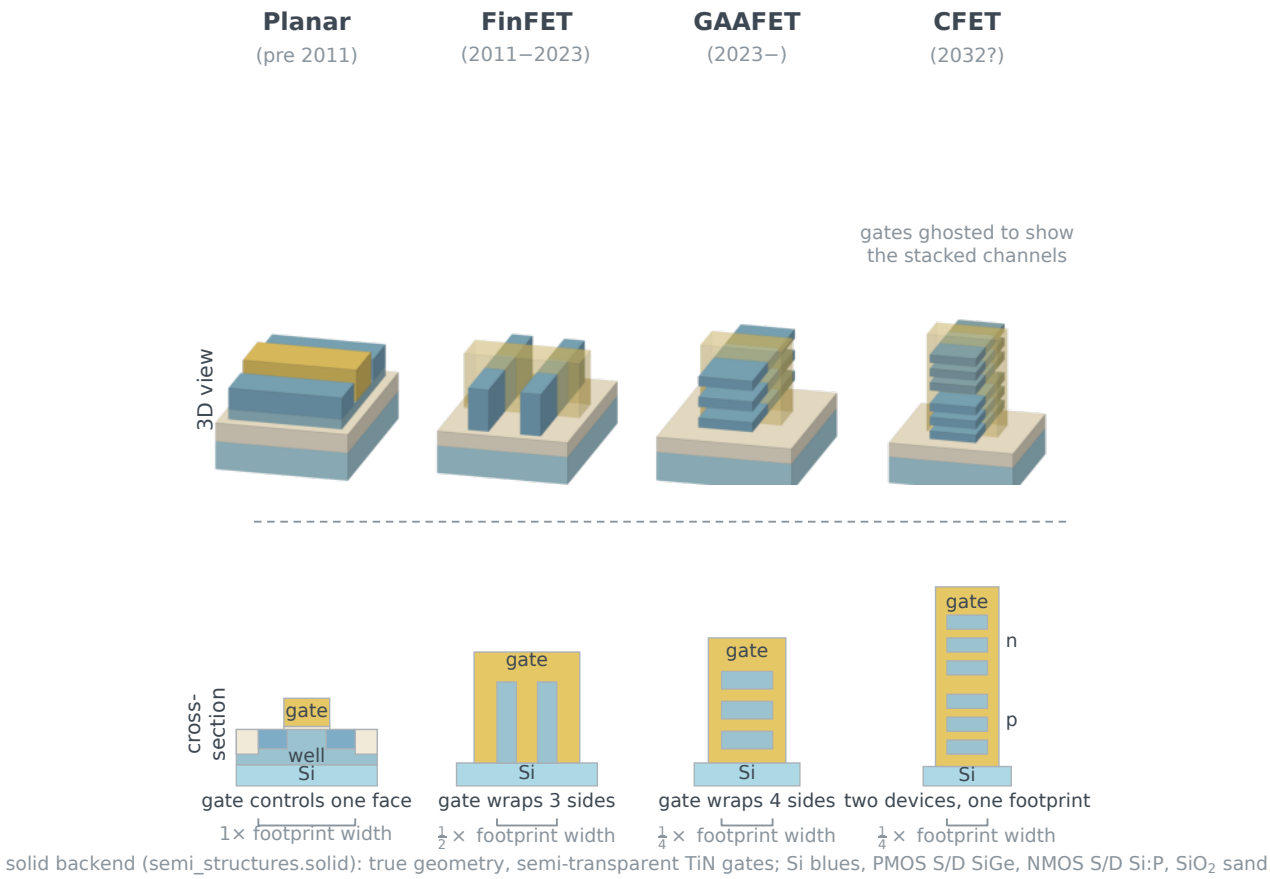
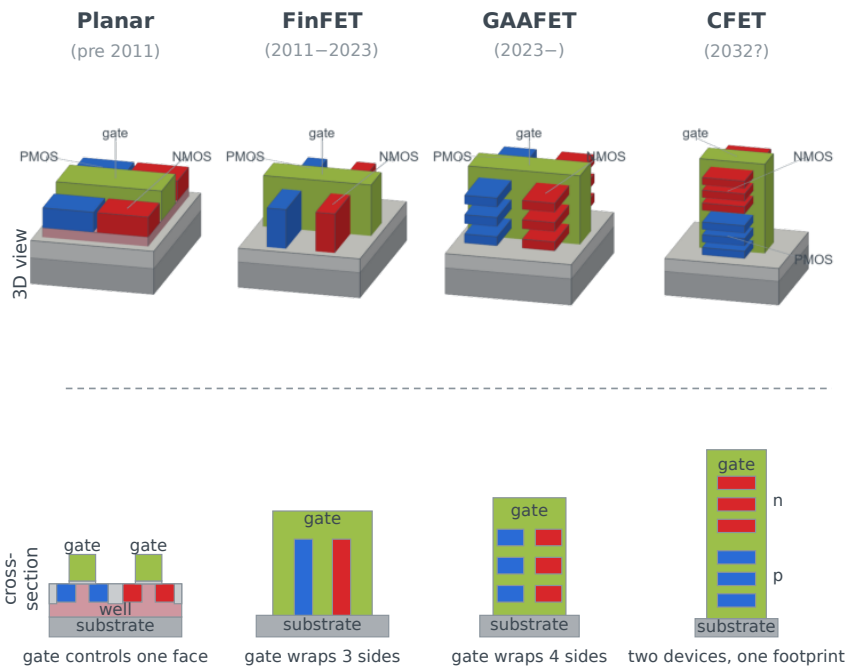


Figure 3: **Special purpose.** The transistor sequence rebuilt with `semi_structures.solid`. The PDF combines rasterized 3D panels with vector labels and cross-sections. The planar section is cut along the channel rather than across it, because the across-channel cut of a planar device shows only a stack of layers. Its active region still spans the footprint bracket, so the widths stay comparable. (Generated with `examples/fig_transistor_solid.py`).



color = circuit role (after the classic evolution diagram): gate green, PMOS blue, NMOS red, well pink, isolation (oxide) and substrate grey; solid backend

Figure 4: Special purpose. The same four devices redrawn in the role colors of the classic evolution diagram (gate green, PMOS blue, NMOS red, well pink, isolation and substrate gray), with a PMOS/NMOS pair in every device, rendered as true solids on the solid backend so fins and nanosheets emerge from opaque gates. The planar section is cut along the channel, which is the only cut that shows a source, a drain and a gate over the channel between them. Color here means circuit role, not material. (Generated with `examples/fig_transistor_evolution_rolecolor.py`).

5 Substrates and orientation features

Every flow in this document starts from a substrate. A circular wafer is a true cylinder rather than a disc drawn to look like one, and its orientation feature is Boolean-cut from the solid, so the notch survives a cutaway and appears correctly in a cross-section.

The cut also reaches every film that inherits the wafer footprint, which is what a blanket deposition does by default. A film given its own footprint is left alone, because it does not extend to the wafer edge. Nothing is deposited in the two panels below, so they show the starting point on its own.

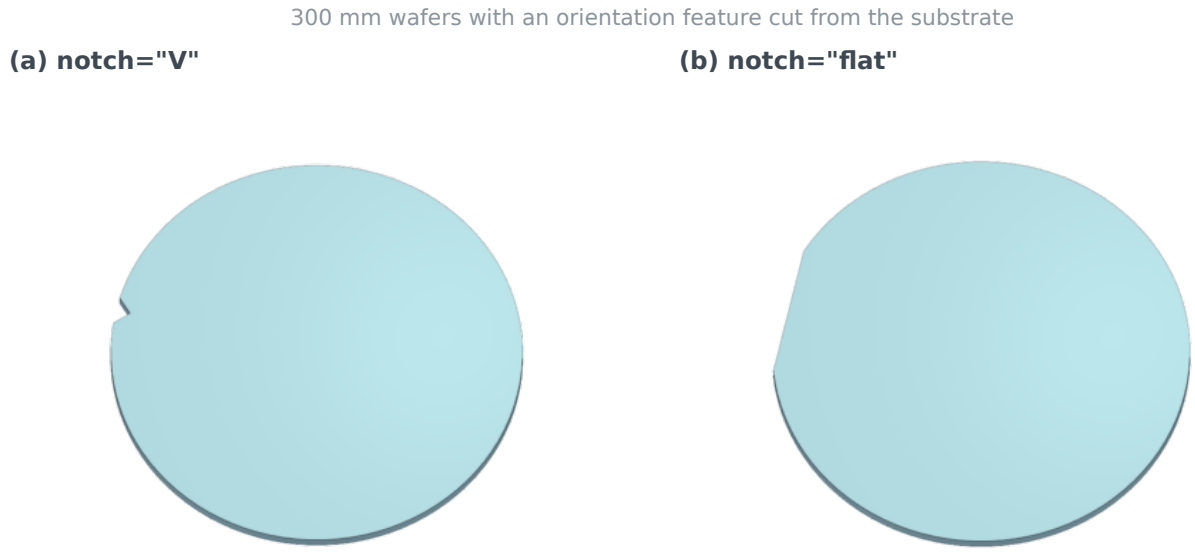


Figure 5: **Process DSL.** Bare 300 mm substrates carrying only their orientation feature: `Wafer(size=300*mm, shape="circle", notch="V")` cuts a 90° V-notch and `notch="flat"` cuts a major flat. `notch_angle` places the feature and `notch_depth` or `flat_length` sizes it, with defaults exaggerated for legibility. The substrate thickness is exaggerated too, because a 775 μm wafer is invisibly thin beside a 300 mm diameter. (Generated with `examples/fig_wafer_notch.py`).

6 Special-purpose figures

Not every useful figure is a process. Film stress bends a wafer, which no flat cross-section can show, so the bow is applied to the slabs themselves and the films are bent with it.

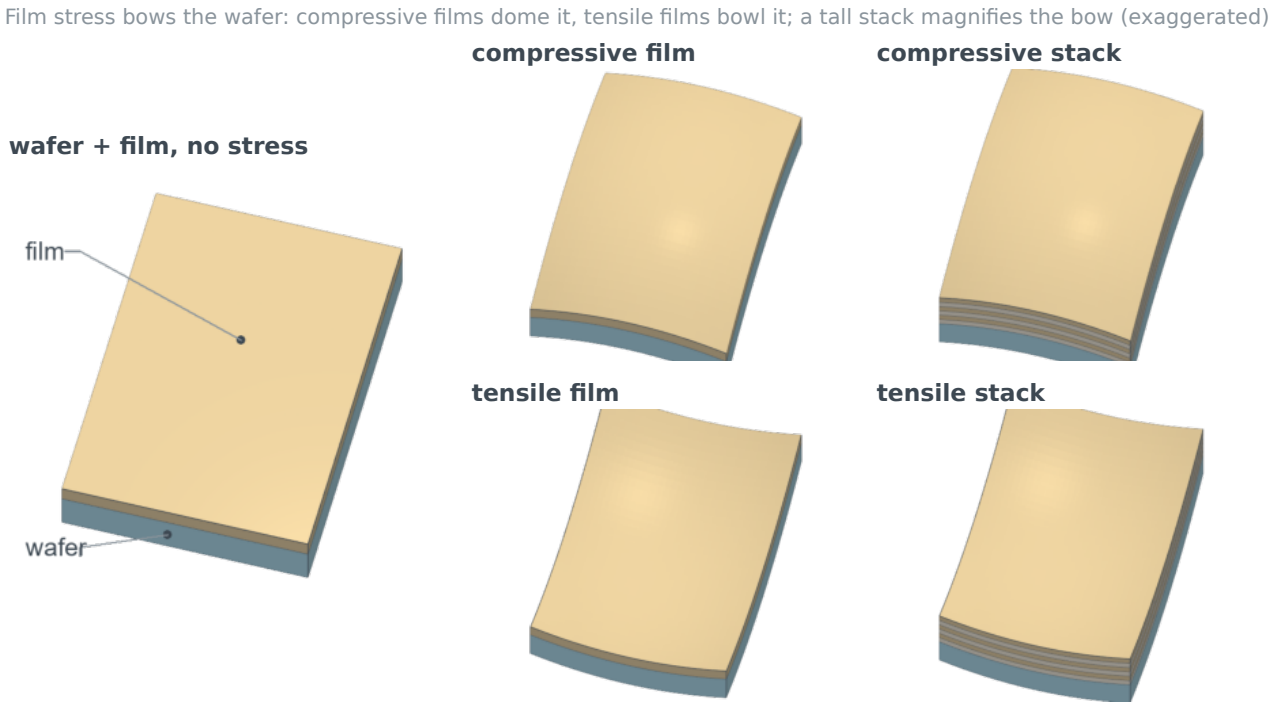


Figure 6: **Special purpose.** Film stress bows the wafer. A flat wafer with one film gives the reference. A compressive film domes the pair and a tensile film bowls it, and a multilayer (an ON stack) magnifies the bow. `Scene.box(..., bow=(sx, sy))` bends every slab with the same displacement, so films stay conformal. Bows are exaggerated and callouts are stick-and-ball (`marker="dot"`). (Generated with `examples/fig_film_stress_bow.py`).

7 The process language in use

A structure written as a fabrication sequence reads like a run sheet, and the renderer follows from the geometry rather than from a decision taken in advance. This section works through that language in order, from the individual verbs to a finished flow written to a file.

The same structure can be specified as a readable fabrication sequence. Six snapshots expose substrate construction, blanket and repeated deposition, depth-controlled rectangle, circle and rotated-ellipse etches, exact-shape fill with optional overfill, and a patterned top layer. The three openings share one line at a constant pitch and the top layer is a 1D grating of equal lines, so the panels show a deliberate layout rather than scattered test shapes.

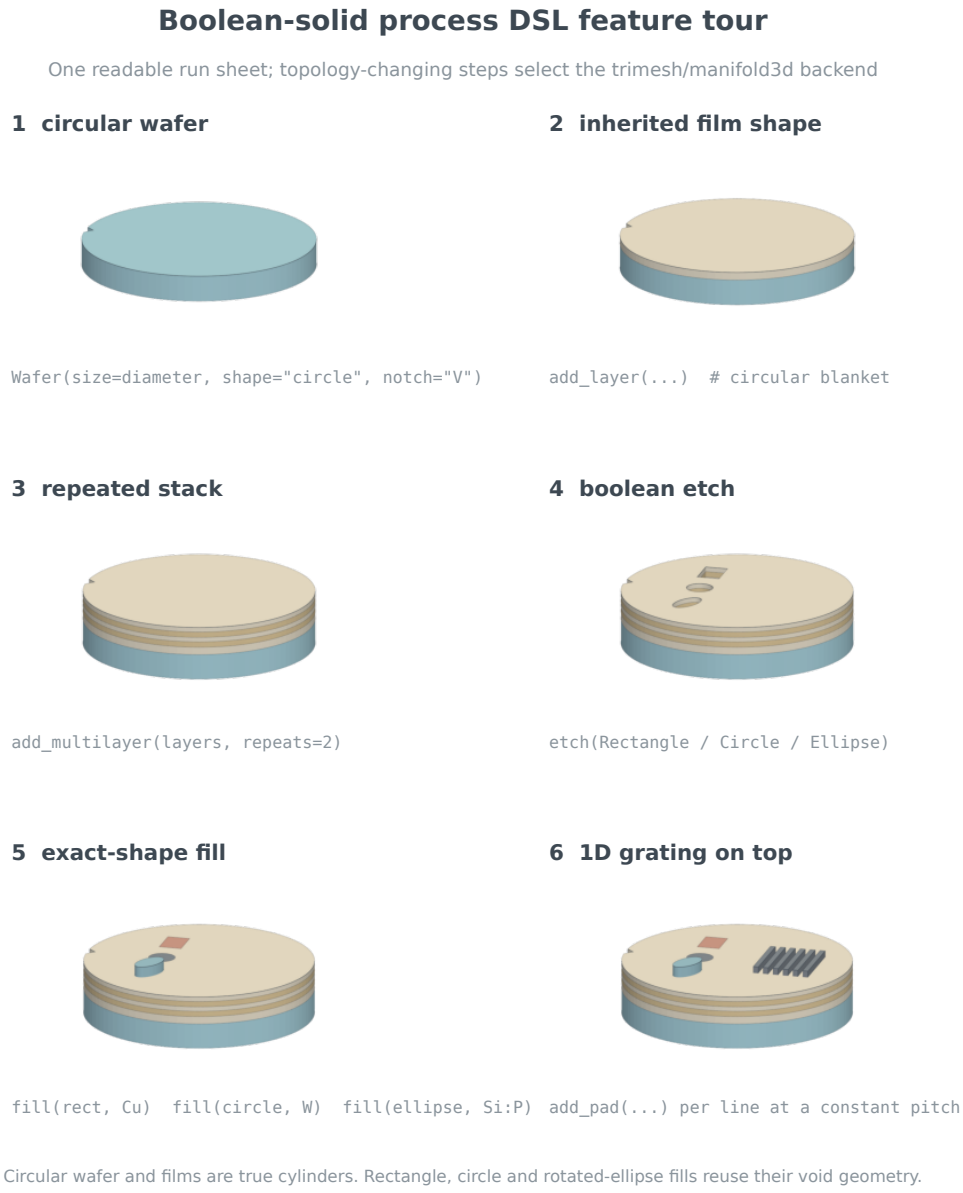


Figure 7: Process DSL. A process-DSL regression example rendered with trimesh/manifold3d booleans and PyVista. Each panel is one circular `Wafer` state. The three etched openings keep genuinely different geometries while sharing one line, and their fills reuse the void shapes exactly. Step 6 adds a 1D grating of six equal lines at a constant pitch, which is what a patterned top layer usually is. (Generated with `examples/fig_dsl_feature_tour.py`).

Shapes are shared between operations rather than restated. One `Rectangle`, `Circle` or `Ellipse` object can drive a film footprint, an etch and the fill that lands in it, so a rotated opening and its contents cannot drift apart. The planar prefix of such a flow is eligible for the vector renderer on a rectangular die, while a full-wafer example selects the solid backend immediately, because its substrate and every inherited film are circular cylinders.

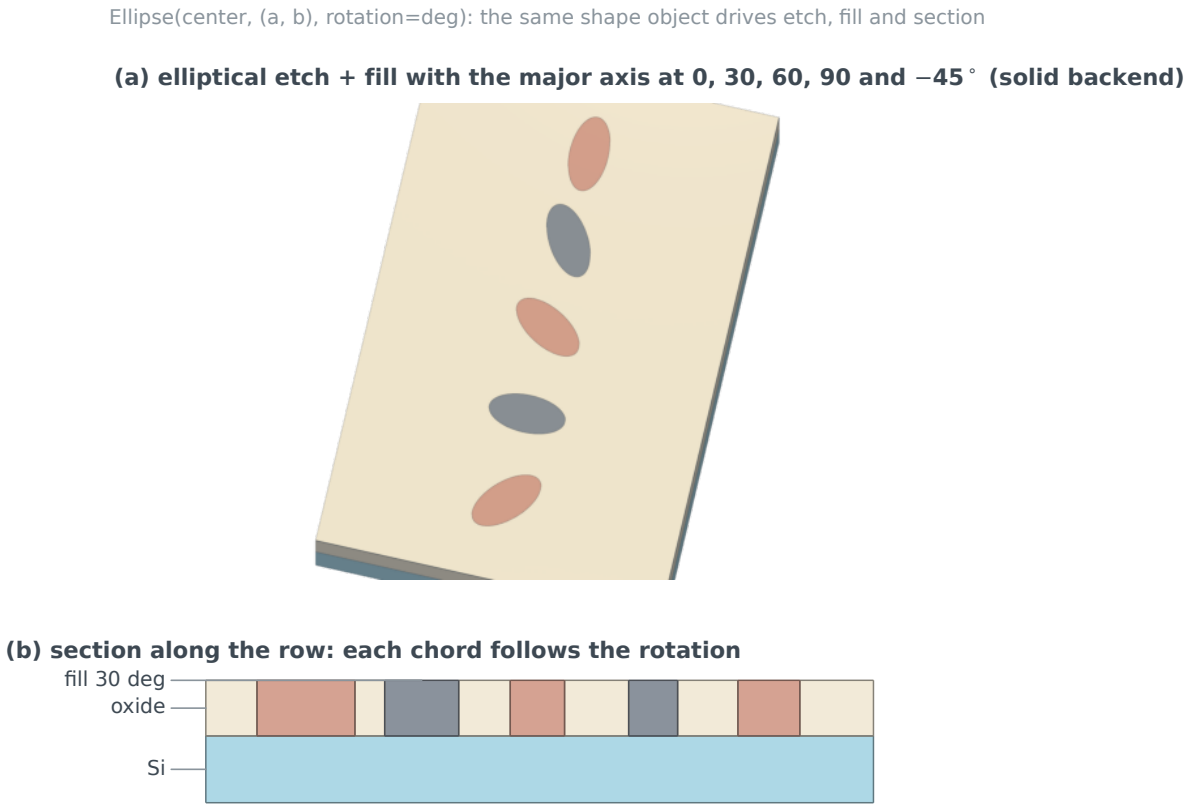


Figure 8: **Process DSL.** Rotated shapes: five elliptical openings etched to the Si with the major axis at 0, 30, 60, 90 and −45° (`Ellipse(center, (a, b), rotation=deg)`) and filled with Cu and W. One shape object drives the exact rotated prisms of the solid backend (a) and the chords of the 2D section along the row (b). (Generated with `examples/fig_rotated_ellipses.py`).

Nesting those two verbs produces the structures that motivated them. A circular etch opens the deposited word-line stack, and progressively smaller etch and fill pairs then form blocking oxide, charge-trap nitride, tunnel oxide, poly-Si channel and oxide core as non-overlapping Boolean solids.

Because every shell is a real solid, the model can be cut open. The cutaway subtracts a camera-facing corner from each part separately, so the exposed faces are capped in their own material colors and the concentric fills read in section. The planes sit on the last hole column before the oxide slit, since a cut at the slit itself would present a blank wall.

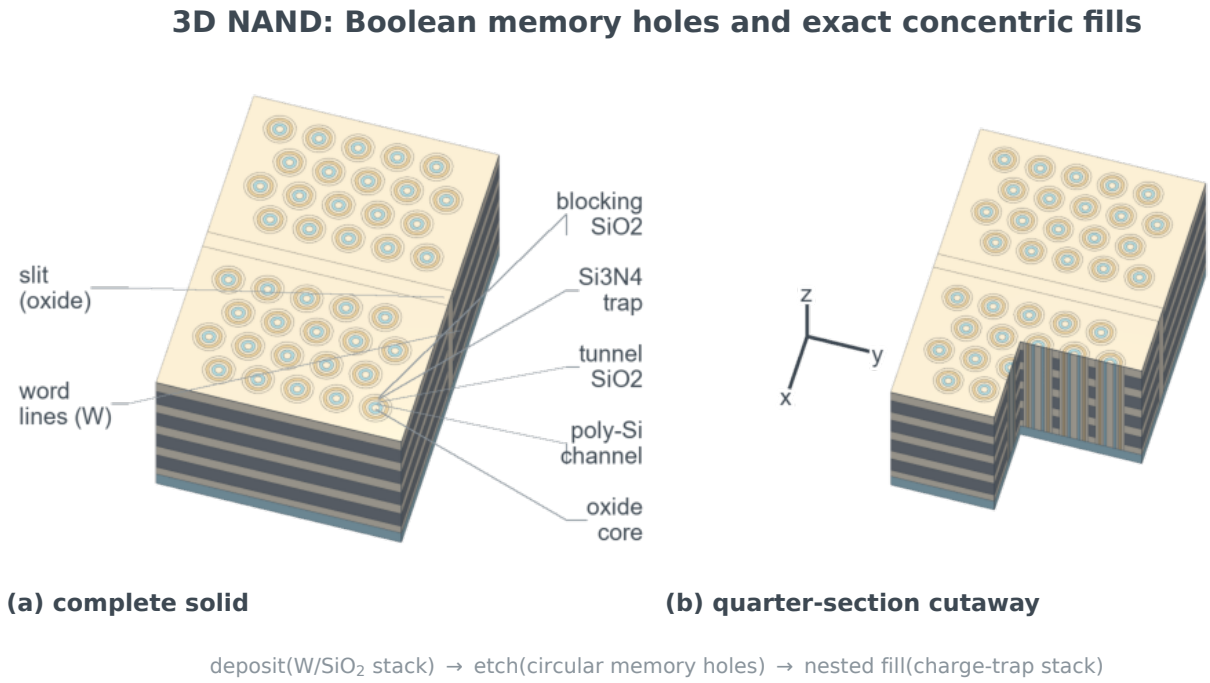


Figure 9: **Process DSL.** A process-DSL 3D-NAND diagram rendered with the Boolean solid backend: a hexagonal (close-packed) memory-hole array in two blocks separated by an oxide-filled slit, as in plan-view micrographs of real 3D NAND, with nested concentric charge-trap fills in every hole (`hex_lattice()` supplies the centers). Panel (b) cuts the same model open at the same size and carries a labelled triad from `Scene.axes()`. Two-line callouts keep the two panels equal. (Generated with `examples/fig_nand3d.py`).

Real stacks are taller than a page. A hundred repeated pairs can be deposited and then drawn as the bottom ten and the top five, with everything that crosses the elided interval cut there by every renderer, so the elision is a property of the model rather than a trick played on one picture.

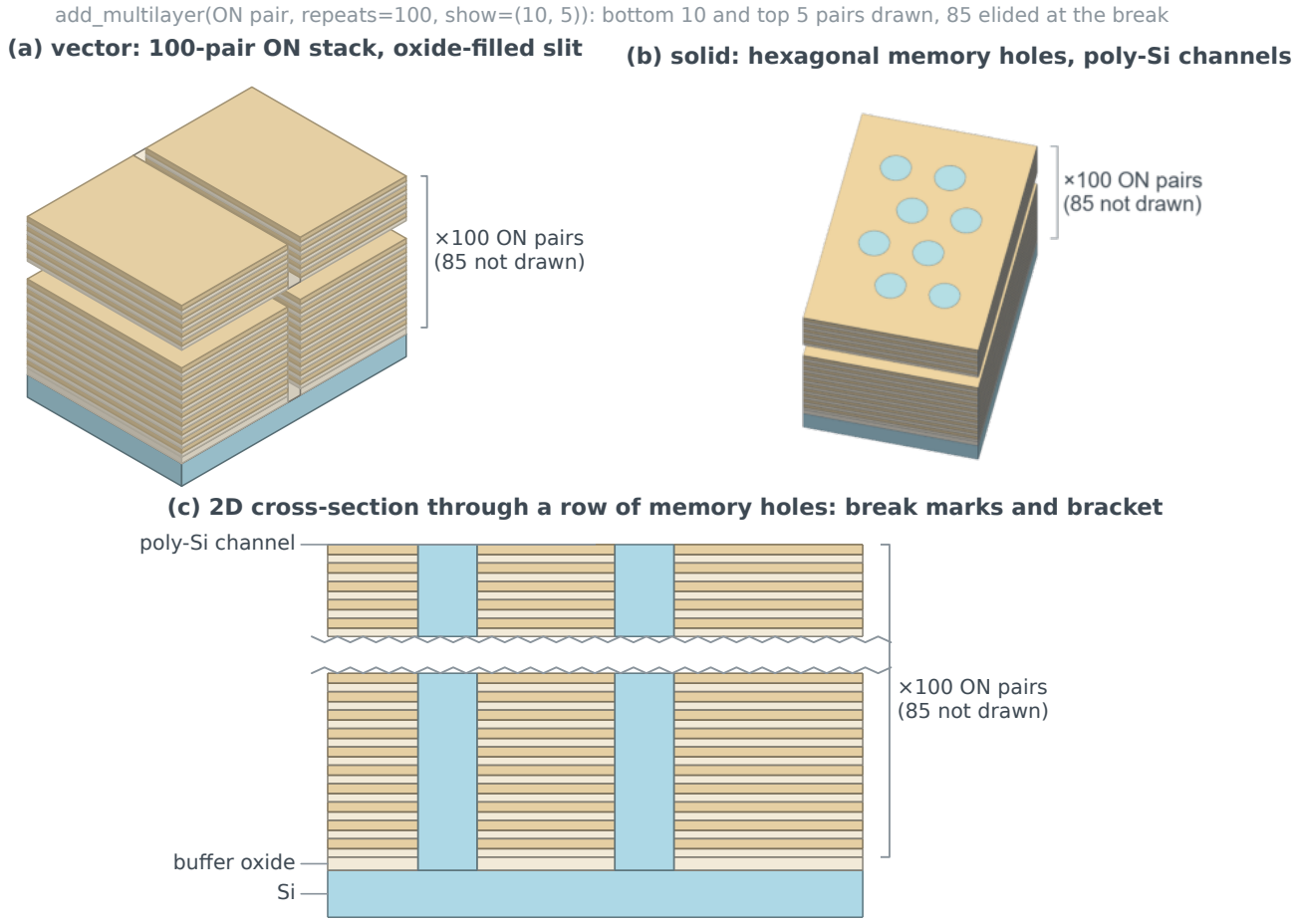


Figure 10: **Process DSL.** Tall repeats without the clutter: a 100-pair SiO_2/Si_3N_4 stack on Si drawn as its bottom ten and top five pairs with a stack break (`add_multilayer(..., repeats=100, show=(10, 5))`). Everything crossing the break is cut there by both 3D renderers. That includes the films, the oxide-filled slit in (a) and the hexagonal memory holes and channels in (b), and (c) is the 2D cross-section of the same model from the new section renderer (`render(view="section")`), with zigzag break marks. every panel brackets the block "×100 ON pairs (85 not drawn)". (Generated with `examples/fig_nand_stack_break.py`).

A flow is not always a single picture. The same model can be captured after each step and drawn as a row of cross-sections, which is how a process is usually explained on a slide or in a lecture.

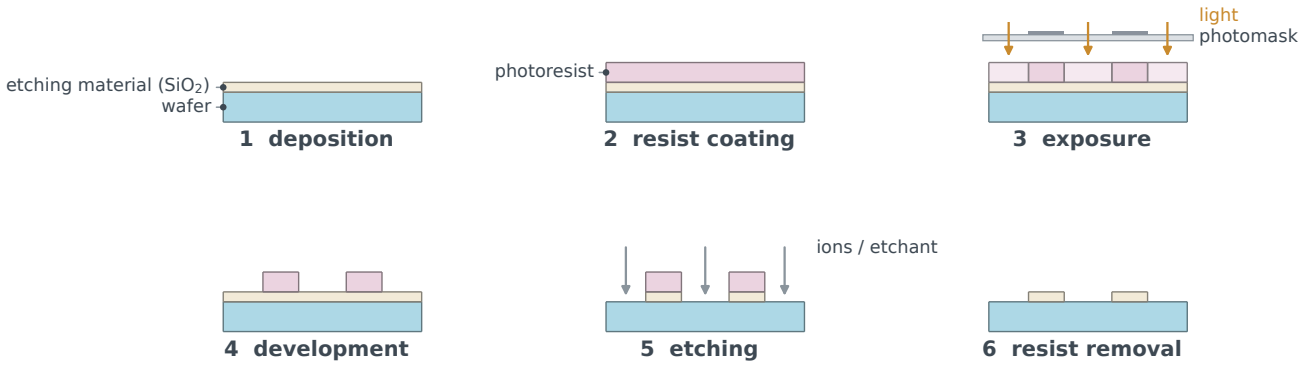


Figure 11: Process DSL. A process flow as a sequence of sections: photolithography in six `Wafer` states drawn by the 2D section renderer. The steps are deposition, resist coating, exposure through a photomask, development and film etch as rectangular etches through the same openings and resist removal with the `strip()` verb. Exposure is a material replacement in the resist. The mask, the light and the ions are annotations, and the light uses the reserved beam color. Callouts are stick-and-ball. (Generated with `examples/fig_litho_flow.py`).

The same six states can be drawn as solids instead. A section says what the layers are, while the 3D view says what the pattern is, and in this case it shows that the openings are trenches running the width of the die rather than isolated holes.

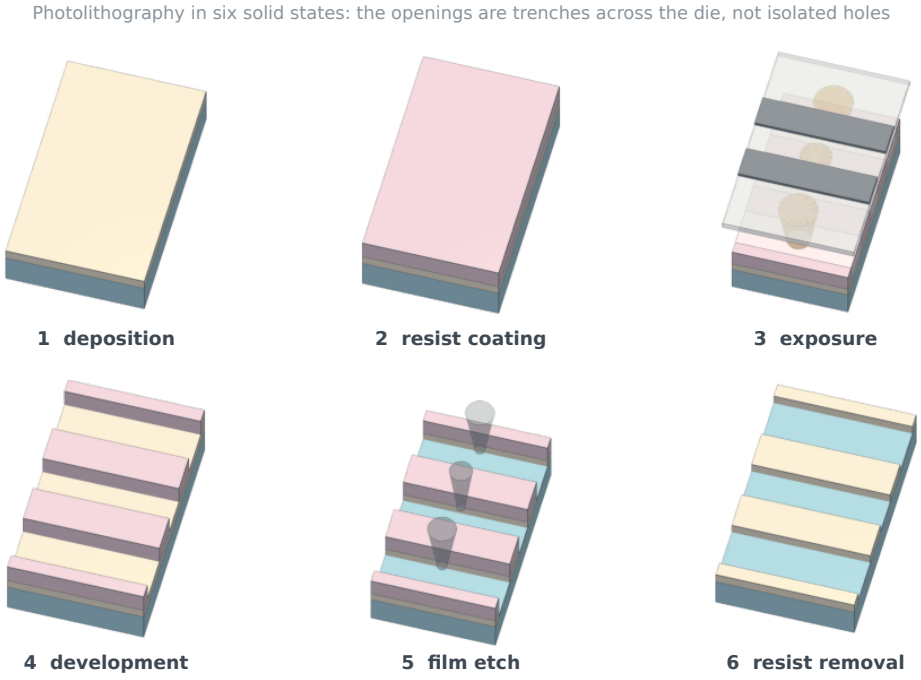


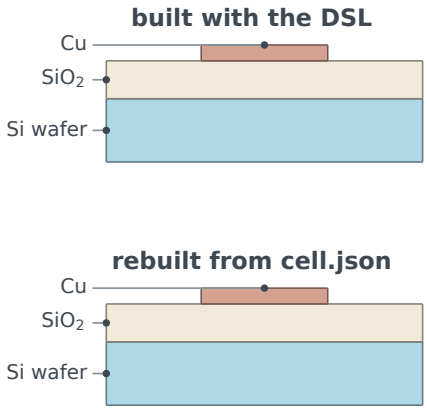
Figure 12: Process DSL. The photolithography flow of the previous figure rendered by the solid backend, one panel per `Wafer` state. The photomask, the exposure light and the ions are annotations rather than process steps, and the beams are tapered `Scene.cone()` solids in the reserved beam color. (Generated with `examples/fig_litho_flow_3d.py`).

Finally, a model can be written down. The document records the ordered calls rather than the geometry they produce, which keeps it short enough to read and exact enough to replay.

A process model round-trips through JSON

cell.json

```
"format": "semi-structures/wafer", "version": 1,
"wafer": {"size": [10.0, 6.0], "thickness": 2.0,
   "name": "Si wafer"},
"operations": [
  {"op": "add_layer", "material": "SiO2",
     "thickness": 1.2, "name": "SiO2"}
  {"op": "add_layer", "material": "Si3N4",
     "thickness": 0.5, "name": "hard mask"}
  {"op": "etch", "shape": {"shape": "rectangle",
     "center": [5.0, 3.0], "size": [4.0, 6.0],
     "rotation": 0.0}, "stop_on": {"$ref": {"op":
     0}}, "name": "trench"}
  {"op": "fill", "hole": {"$ref": {"op": 2}},
     "material": "Cu", "name": "Cu"}
  {"op": "strip", "target": "hard mask"}
]
```



identical pixels: SHA-256 of the two renderings agree

Figure 13: Process DSL. A process model saved as JSON and rebuilt from it. The document stores the *recipe*, which is the ordered DSL calls with their arguments, rather than the derived geometry, so replaying it reconstructs every layer, hole and fill. Arguments left at their defaults are omitted, and feature references such as `stop_on` and `fill` are stored as `$ref` addresses into the operation list, which makes them independent of naming. The two renderings are compared by SHA-256 of their pixels: the reproduced model is not merely similar but the same picture. (Generated with `examples/fig_serialization_roundtrip.py`).

8 Solid-backend labels tied to 3D features

Callouts can now be authored directly on a solid scene. Each anchor is a 3D feature coordinate that follows camera changes, while its label position and optional elbow points use normalized screen coordinates for stable publication layout. Leaders terminate at the text boundary and labels remain horizontal. The supplied visual reference was traced to Hanwha's feature on HBM packaging. The panel below is an independent schematic rather than a reproduction of that image. [3]

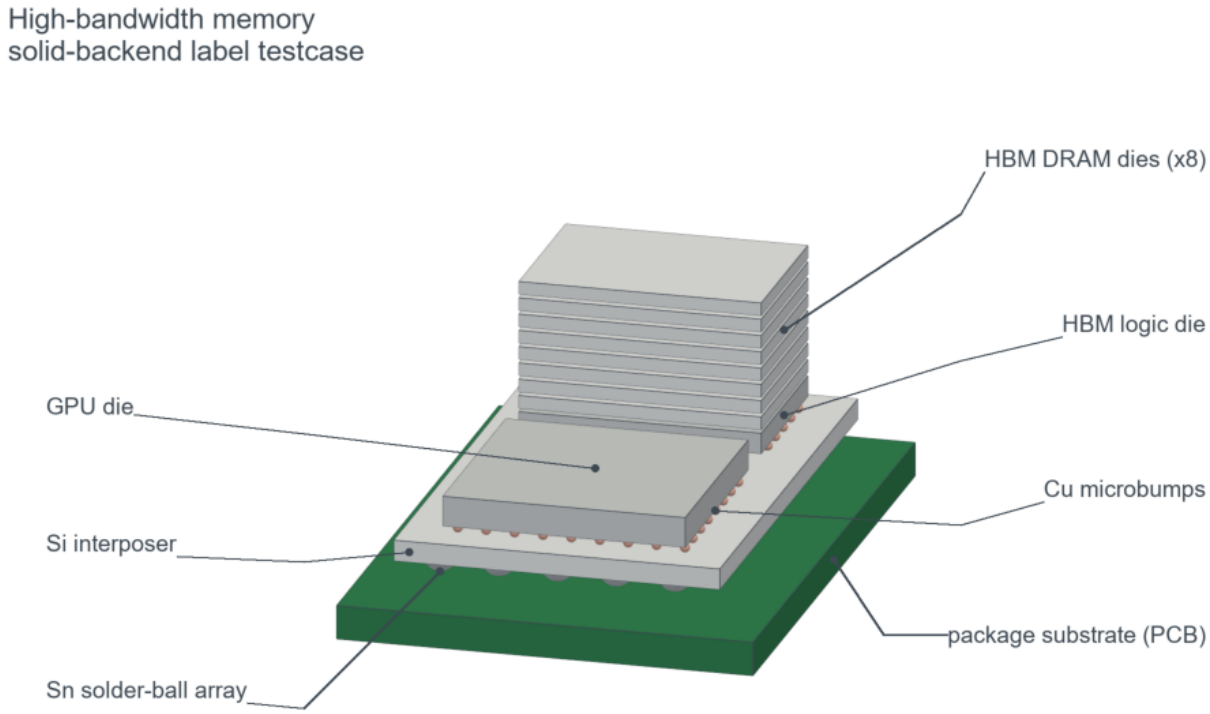


Figure 14: Special purpose. An HBM package testcase. The solid backend creates the PCB package substrate, interposer, GPU, eight-die HBM stack, microbumps and Sn solder-ball array. In this packaging view the silicon parts take the neutral die gray (logic dies one shade darker) so the package materials carry the color. `Scene.label()` supplies every routed callout, sized in points at print width. (Generated with `examples/fig_hbm_labeled.py`).

9 Miscellaneous structures

Three device families that share no process but do share the drawing language: compound-semiconductor emitters, wide-bandgap power devices and a backside power-delivery block.

9.1 III-V and III-nitride emitters

Alloy-aware layer colors make it possible to compare six landmarks in one drawing language. The sequence runs from the GaAs junction laser through the double heterostructure, the DFB and surface-emitting cavities, the InGaN blue LED and a mesa micro-LED. Each alloy fill is interpolated from its endpoint materials, so an AlGaAs cladding sits visibly between AlAs and GaAs rather than taking an arbitrary green. [4, 5, 6, 7, 8, 9]

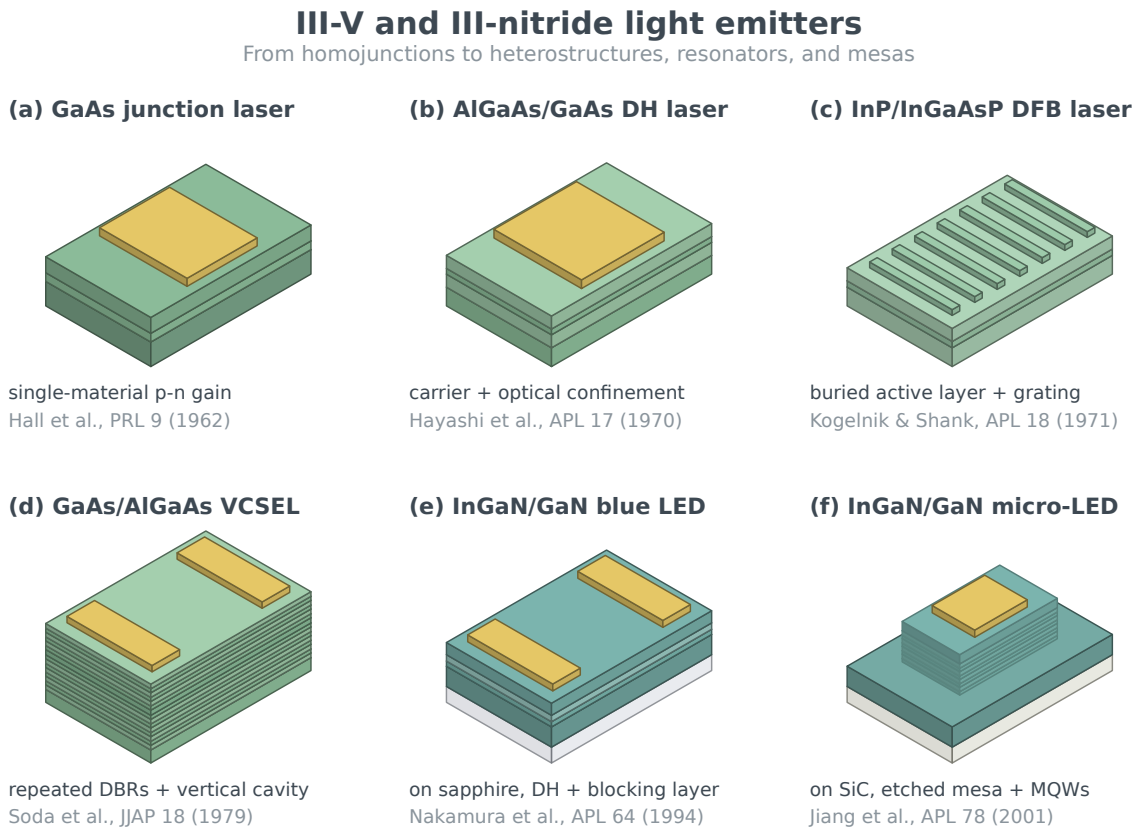


Figure 15: Process DSL. III-V laser and III-nitride LED structures. The micro-LED mesa is a field etch expressed through the process DSL. Because it is rectilinear the vector backend draws it exactly, in the same style as its neighbors. The two nitride emitters sit on transparent substrates, sapphire in (e) and a bulk SiC wafer in (f), drawn translucent by the palette's alpha. Dimensions are schematic and selected for visual comparison. (Generated with `examples/fig_optoelectronic_devices.py`).

9.2 4H-SiC power devices

The power-device page exercises doped shades, vertical drift regions, implanted wells, surface and trench gates, shielding regions and repeated fin channels. Four structures are drawn: a Schottky diode with backside metal, a planar DMOS, a trench MOSFET with nested oxide and gate fills, and a vertical tri-gate MOSFET. Implanted regions keep the SiC hue and move one shade darker, so doping never reads as a different material. [10, 11, 12, 13]

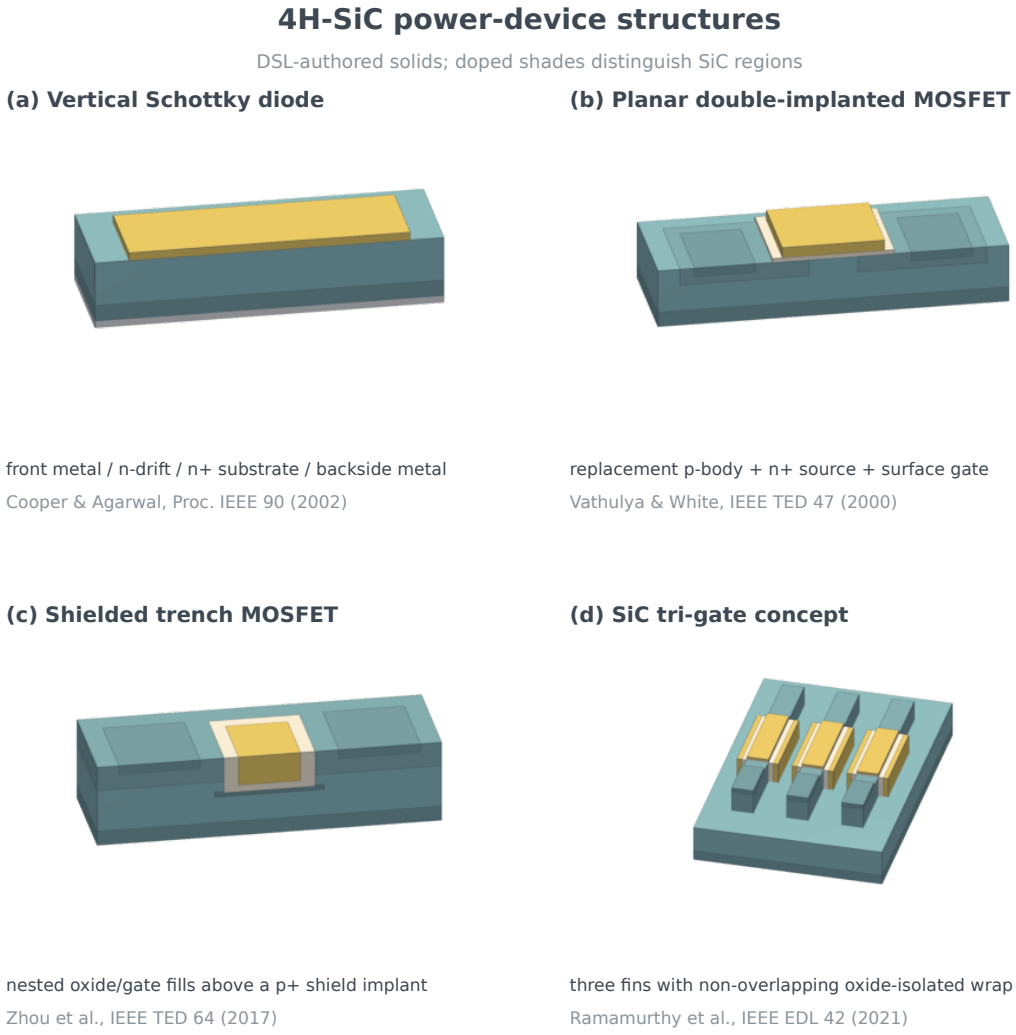


Figure 16: **Process DSL.** Representative 4H-SiC power structures. Implants replace the drift material, and the trench oxide/gate are exact nested fills. Doped regions retain the SiC hue so material identity is not confused with circuit role. (Generated with `examples/fig_sic_power_devices.py`).

9.3 Backside power delivery

Backside power separates coarse power conductors from frontside signal routing and connects them through nano-TSVs and buried rails. The geometry is grounded in published buried-power-rail integrations. [14, 15, 16] The two supplied visual references were traced to Intel's PowerVia Figure 4 and Aminext's nTSV explainer. [17, 18] The rendering below is a new solid-model testcase, not a reproduction of either source.

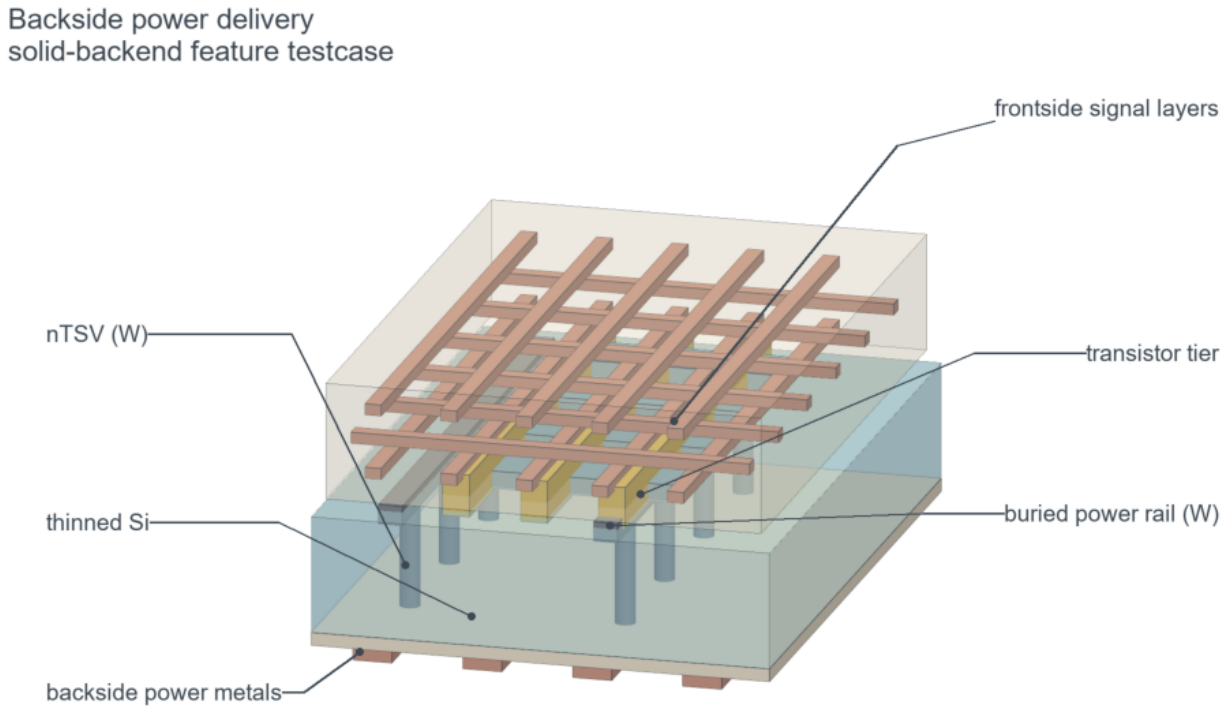


Figure 17: Special purpose. A labeled backside-power stack. Wide backside Cu, W nano-TSVs, buried W rails, a transistor tier and frontside Cu signal layers are separate true-solid objects. Each stick-and-ball callout was checked against the geometry and the rendered pixel, so every dot sits on a visible face of the part it names. (Generated with `examples/fig_backside_power_delivery.py`).

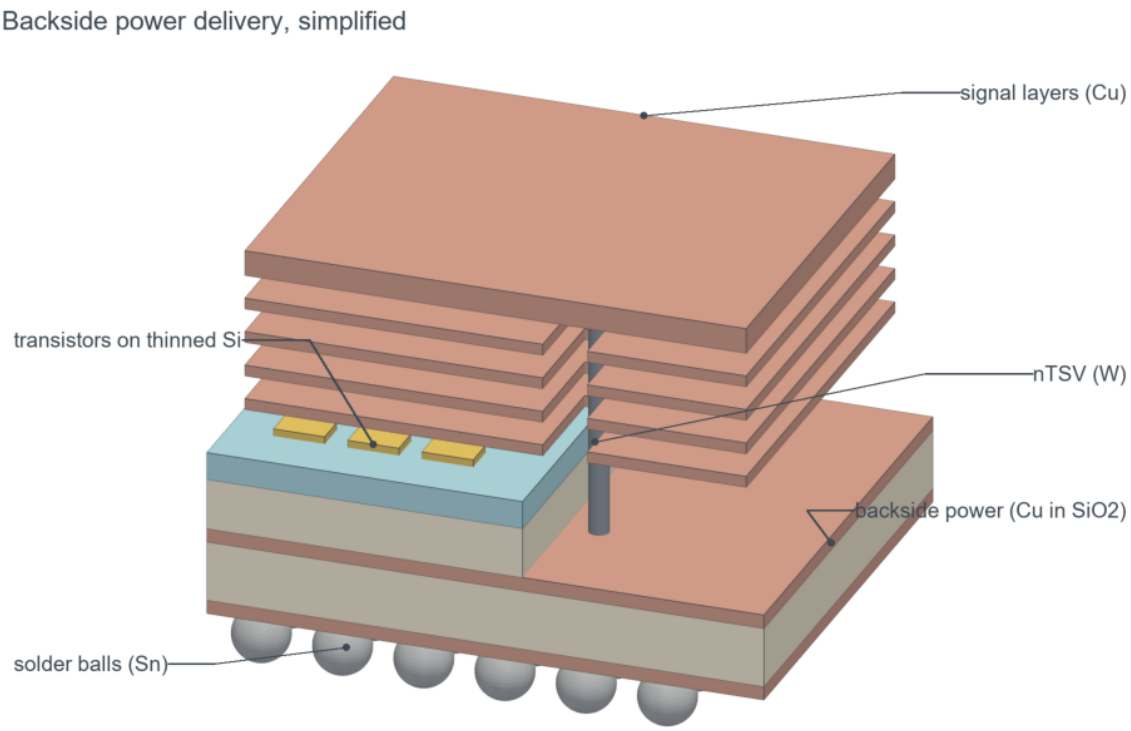


Figure 18: Special purpose. The same idea as a simplified block, authored from a stylized reference illustration: Cu signal layers exploded above as plates, slotted where the single W nano-TSV runs up through them. Below that sit transistor pads on the thinned Si, a stepped backside power block of Cu sheets in SiO_2, and Sn solder balls. Color is material here. The reference's orange-on-dark scheme would be a presentation override. (Generated with `examples/fig_backside_power_simple.py`).

10 Authoring from reference images

The figures in this section were produced the way the introduction describes: a multimodal coding assistant was given a reference image and asked to draw the structure with the package. In each case the image supplied only the composition, viewpoint and callout set. The geometry is new, schematic and built from package primitives, and the reference images are not redistributed (their provenance is recorded in `examples/REFERENCES.md`). The first two figures are the same GPU package, assembled and then exploded, in the package's material palette. The remaining two were requested *in the colors of their references*, as was the role-colored transistor line-up in Section 4: circuit-role or presentation colors (raw hex fills, a dark background) are permitted by both renderers, but they are deliberate exceptions to the palette law and are kept as separate scripts so the material-colored figures remain the defaults.

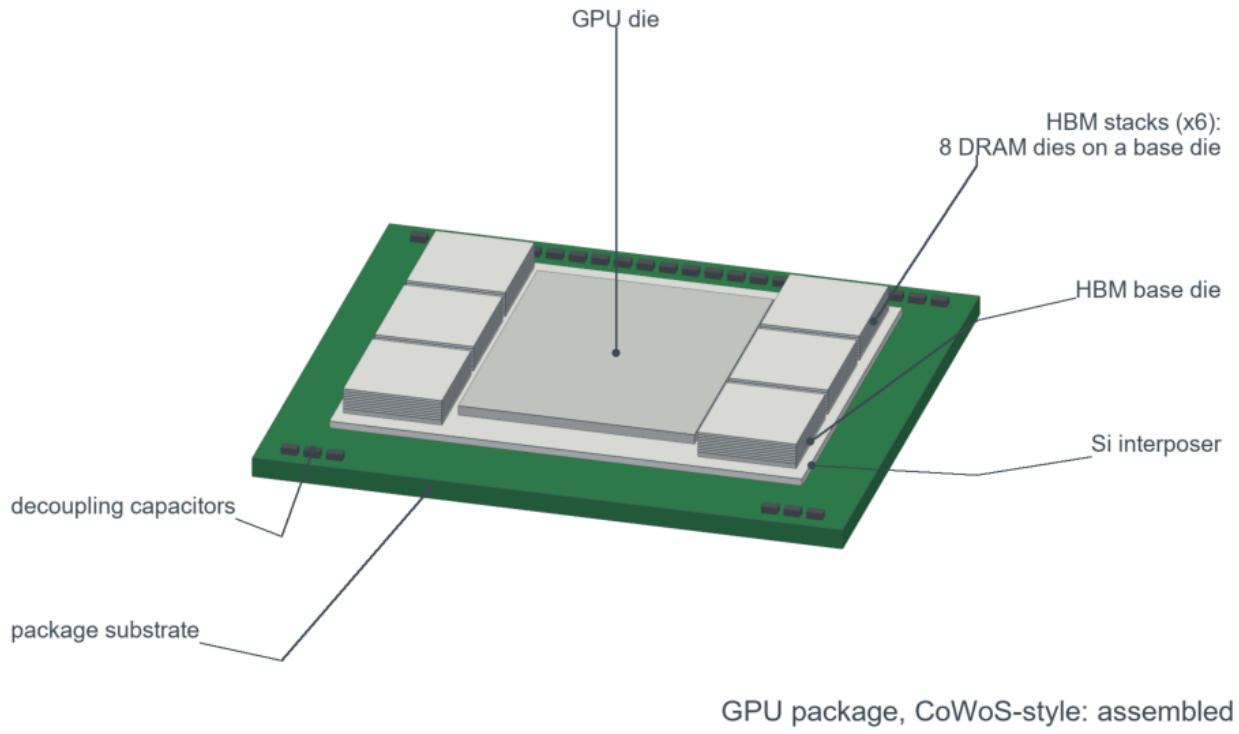


Figure 19: Special purpose. A modern GPU package assembled, authored from a package photograph: the organic package substrate and its decoupling capacitors, the silicon interposer seated on its C4 bump field, and on top the GPU die flanked by six HBM towers, three per side. Assembled, both bump fields are hidden, so the callouts name only what the camera can see. Each is a stick-and-ball leader from `Scene.label()`, and every anchor was checked against the geometry and the rendered pixel. (Generated with `examples/fig_gpu_cowos_assembled.py`).

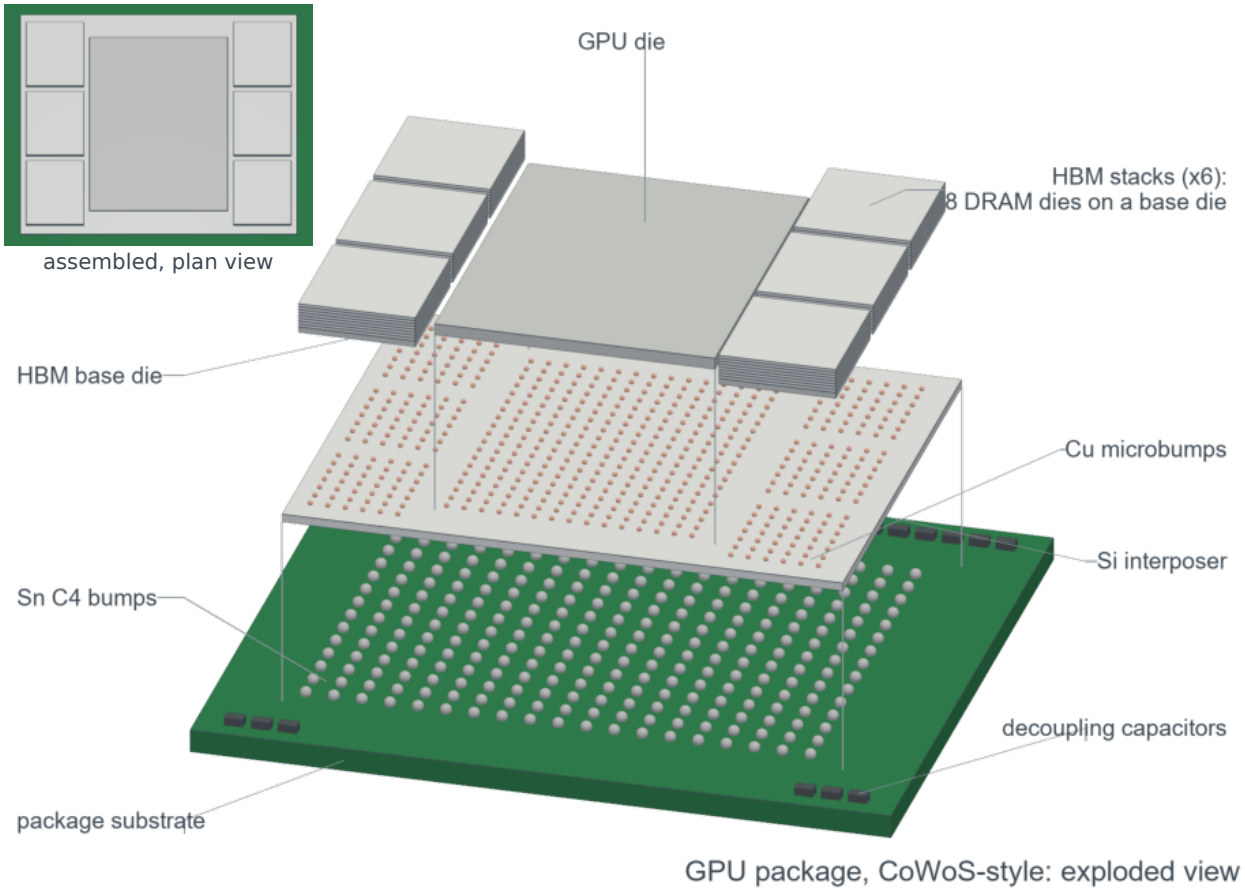


Figure 20: Special purpose. A modern GPU package as an exploded chip-on-wafer-on-substrate (CoWoS-style) view, authored from a package photograph and a foundry exploded-view graphic: package substrate with its Sn C4 bump field and decoupling capacitors, Si interposer with Cu microbump fields under every die, and on top the GPU die flanked by six HBM towers (eight DRAM dies on a base die with Cu bond layers, three per side). Thin corner guide lines tie each level to the one below, and the inset repeats the previous figure from almost directly above, the plan view of the photograph. Silicon parts take the neutral die gray with the logic dies one shade darker, so PCB green, Sn and Cu carry the color. (Generated with `examples/fig_gpu_cowos_exploded.py`).

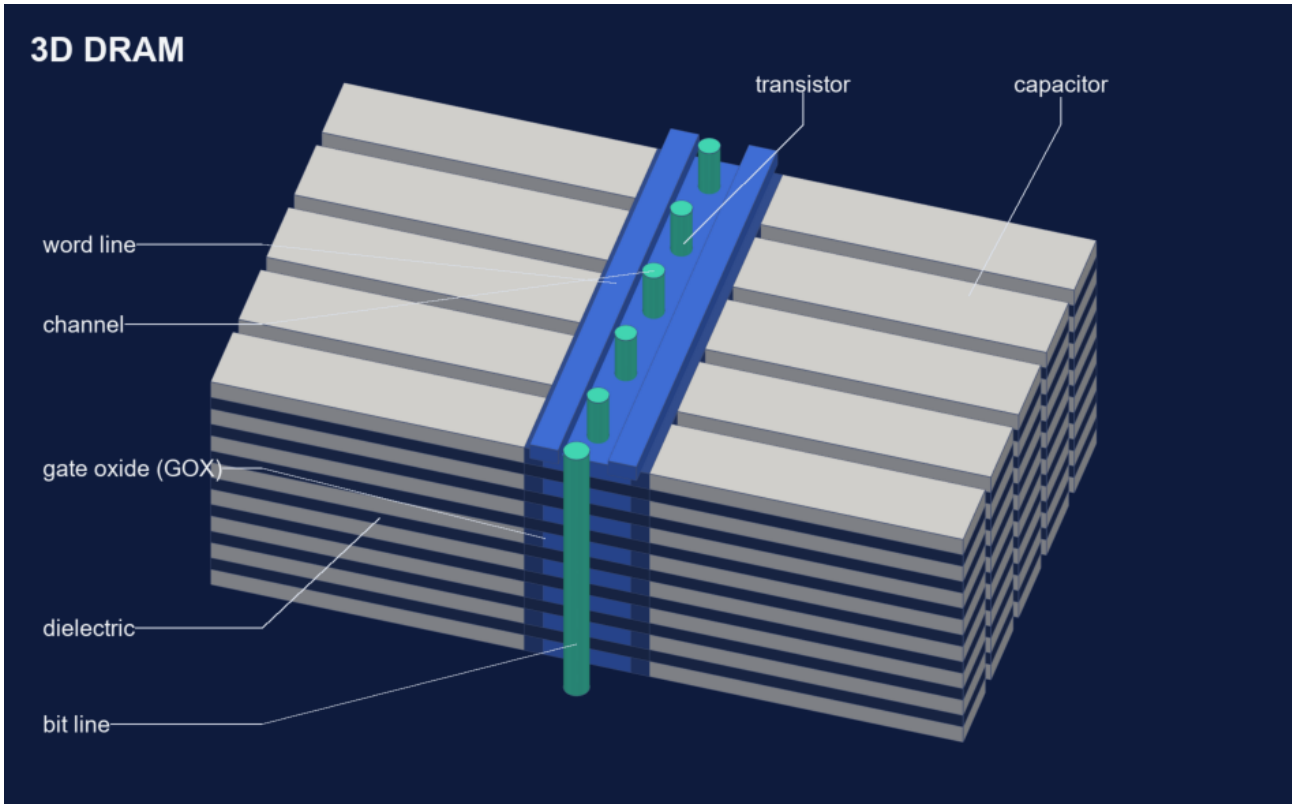


Figure 21: Special purpose. A horizontal-capacitor 3D DRAM archetype in a vendor-style presentation scheme (dark background, light-gray capacitor plates, dark dielectric, blue word lines and transistors, teal channels and bit line, light callouts) using `render(background=..., edge_color=...)`. (Generated with `examples/fig_dram3d_reference_colors.py`).

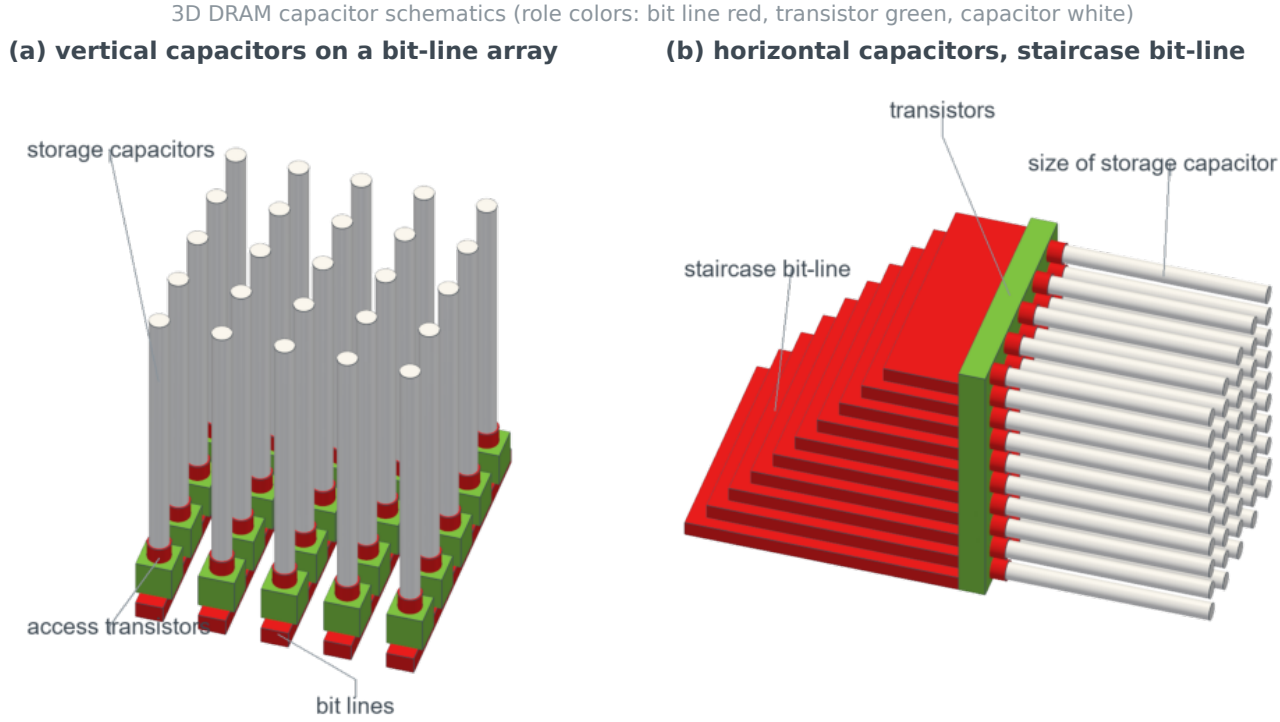


Figure 22: Special purpose. Two 3D DRAM capacitor schematics in a red/green/white role scheme: (a) vertical storage capacitors on access transistors over a bit-line array. (b) Horizontal capacitors on a staircase bit-line, the capacitor length setting the storage size. Horizontal cylinders are rotated trimesh primitives added with `Scene.add()`. (Generated with `examples/fig_dram3d_capacitor_schematics.py`).

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